

OSAZEE ERO, PhD

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AI & MACHINE LEARNING ENGINEER

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with experience in engineering simulation, industrial computer vision, and LLM-assisted workflows. PhD in Mechanical and Mechatronics Engineering from the University of Waterloo. Builds practical AI applications, including a deployed document assistant with hybrid retrieval and source citations, alongside applied ML research and modeling.

TECHNICAL SKILLS

- **ML & data:** Python, SQL, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, XGBoost; classification, regression, computer vision, image segmentation.
- **Generative AI:** RAG, LLM evaluation, tool calling, agentic workflows, LangChain, LlamaIndex, OpenAI API, vector databases, LoRA/QLoRA.
- **Application delivery:** FastAPI, REST APIs, PostgreSQL/pgvector, Docker, AWS ECS, S3, MLflow, Git, Next.js, TypeScript.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer/Researcher

Phaseshift Technologies | Oct 2024 - Present

- Contribute to ML models for accelerating engineering simulation workflows and supporting materials prediction.
- Build and evaluate classification and regression models using engineering and simulation data.
- Develop LLM-assisted workflows for simulation code generation and parameter configuration.
- Contribute to agentic workflows connecting ML predictors, simulation tools, and optimization steps for materials exploration.

Machine Learning Research Associate

Multi-Scale Additive Manufacturing Lab, University of Waterloo | Sep 2024 - Oct 2025

- Developed computer vision and deep learning models for identifying process deviations in manufacturing image and video data.
- Prepared image, video, and sensor data for training and evaluation; explored inference optimizations for real-time monitoring.

Lead Software Developer

OptiFab | Jan 2023 - Jun 2024

- Developed Python-based 3D simulation tools for laser scan-path optimization in additive manufacturing.
- Translated manufacturing requirements into software features for simulating and comparing scanning strategies.

PhD Researcher - Machine Learning & Computer Vision

University of Waterloo | Sep 2020 - Jul 2024

- Investigated self-organizing maps and U-Net segmentation for detecting defect-related patterns in optical tomography images from laser powder bed fusion.
- Combined fuzzy logic and machine learning for porosity detection, evaluated findings against computed tomography measurements, and published peer-reviewed research.

SELECTED PERSONAL PROJECTS

Contextly | Deployed document intelligence application

contextly.osazeeero.com | [Source code](#)

- Built a PDF question-answering application with page-aware ingestion, hybrid semantic and keyword retrieval, and document/page citations.
- Implemented authenticated document access, persistent conversations, upload failure reporting, and retry workflows using Next.js, FastAPI, and PostgreSQL/pgvector.
- Deployed the frontend on Vercel and a Dockerized API on AWS ECS Fargate, with RDS and private S3 storage; added retrieval evaluation and regression tests.

Hand-Controlled Virtual Lab | Computer vision prototype

[Demo and project details](#) | [Source repository](#)

- Co-developed a prototype connecting OAK-D hand detection to a Unity virtual laboratory, using TensorFlow and Python with TCP communication for basic hand-position control.

ADDITIONAL TEACHING EXPERIENCE

Teaching Assistant

University of Waterloo | May 2021 - Oct 2024

- Supported MTE 262 and MTE 325 labs in digital logic, PLC programming, and microprocessor systems; guided students through programming and hardware interfacing exercises.

University Lecturer

University of Benin | Jul 2013 - Aug 2020

- Taught electrical and electronic engineering courses, designed practical assignments, and supervised undergraduate projects involving programming and engineering systems.

EDUCATION

PhD, Mechanical and Mechatronics Engineering | University of Waterloo | 2024

MSc, Systems Engineering | University of Lagos | 2016

BEng, Electrical/Electronic Engineering | University of Benin | 2011

TRAINING

AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents | Udemy | Completed

Machine Learning Engineer Nanodegree | Udacity | Completed, 2021

In progress (Udemy): AI Engineer Agentic Track: The Complete Agent & MCP Course; AI Engineer Production Track: Deploy LLMs & Agents at Scale.

SELECTED PUBLICATIONS

[Optical tomography and machine learning for in-situ defects detection in laser powder bed fusion: A self-organizing map and U-Net based approach](#). Additive Manufacturing, 2023.

[An integrated fuzzy logic and machine learning platform for porosity detection using optical tomography imaging during laser powder bed fusion](#). International Journal of Extreme Manufacturing, 2024.